

USE-CASE FLOW

Use Case ID: UC-01

Use Case Name: Monitor Microservice Health

Primary Actor: DevOps Engineer

Supporting Actor: Microservice

Priority: High

1. Preconditions

1. The microservice is registered in the service catalog.
2. A health endpoint is available for the microservice.
3. The health-check monitoring process is active.
4. The DevOps Engineer has access to the portal.

2. Main Success Scenario

1. The system starts the health-check cycle.
2. The system sends a health probe to the microservice health endpoint.
3. The microservice responds successfully to the health probe.
4. The system records the successful health-check result.
5. The system updates the service availability information.
6. The system updates the rolling 24-hour service availability percentage.
7. The system waits 30 seconds before the next health probe.
8. The DevOps Engineer can view the current health status of the microservice.

3. Alternate Flow

- 3a. Microservice does not respond to the health probe.
 - 3a1. The system records the failed health probe.
 - 3a2. The system waits 30 seconds and sends the next health probe.
 - 3a3. If three consecutive health probes fail, the system records the microservice as down.
 - 3a4. The system records the downtime event.
 - 3a5. The system dispatches a downtime alert to the DevOps Engineer.
 - 3a6. The DevOps Engineer investigates the affected microservice.

4. Postconditions

1. The latest health-check result is recorded.
2. The rolling 24-hour availability percentage is updated.
3. If three consecutive probes fail, downtime is recorded and an alert is dispatched.